S.No: 1	Exp. Name: ***Registration Form using Table***	**Date: 2026-08-15**

Aim:
Build a registration form laid out inside an HTML table.

Form Fields

- **Name**: A text input with `name="name"`.
- **Password**: A password input with `name="password"`.
- **E-mail**: An email input with `name="email"`.
- **Phone Number**: A telephone input with `type="tel"` and `name="phone"`.
- **Sex**: Two radio buttons for "Male" and "Female", both with `name="sex"`.
- **Date of Birth**: Three dropdowns with `name="day"`, `name="month"`, and `name="year"`.
- **Languages Known**: Four checkboxes for English, Telugu, Hindi, and Tamil, each with a matching lowercase `name` attribute.
- **Address**: A textarea with `name="address"`.

Table Layout

- Place each label in one column and its input in the adjacent column.
- Set the table border to `1`.

Buttons

- A <button> element with `type="submit"` and the text "Submit".
- A <button> element with `type="reset"` and the text "Reset".

The form should look like this.

ID: A24126552168
2024-2028-CSM-C
ANIL NEERUKONDA INSTITUTE OF TECHNOLOGY & SCIENCES

Registration Form

Name:	
Password:	
E-mail:	
Phone Number:	
Sex:	○ Male ○ Female
Date of Birth:	Day ▾ Month ▾ Year ▾
Languages Known:	☐ English ☐ Telugu ☐ Hindi ☐ Tamil
Address:	
Submit Reset	

Source Code:

index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-
scale=1.0">
    <title>Registration Form</title>
</head>
<body>
<h2>Registration Form</h2>
<!--Write your code here...-->

<form>
        <table border="1">
                <tr>
                        <td>Name:</td>
                        <td><input type="text" name="name"></td>
                </tr>
                <tr>
                        <td>Password:</td>
                        <td><input type="password" name="password">
</td>
                </tr>

                <tr>
                        <td>E-mail:</td>
                        <td><input type="email" name="email"></td>
                </tr>

                <tr>
                        <td>Phone Number:</td>
                        <td><input type="tel" name="phone"></td>
                </tr>

                <tr>
                        <td>Sex:</td>
                        <td>
                                <input type="radio" name="sex"
value="Male">Male<br>
                        <input type="radio" name="sex"
value="Female">Female
                        </td>
                </tr>
```

```
                <tr>
                        <td>Date Of Birth:</td>
                        <td>
                                <select name="day">
                                        <option>Day</option>
                                        <option>1</option>
                                        <option>2</option>
                                        <option>3</option>
                                        <option>4</option>
                                        <option>5</option>
                                </select>

                                        <select name="month">
                                                <option>Month</option>
                                                <option>Jan</option>
                                                <option>Feb</option>
                                                <option>Mar</option>
                                                <option>Apr</option>
                                                <option>May</option>
                                        </select>

                                        <select name="year">
                                                <option>Year</option>
                                                <option>2000</option>
                                                <option>2001</option>
                                                <option>2002</option>
                                                <option>2003</option>
                                                <option>2004</option>
                                                <option>2005</option>
                                                <option>2006</option>
                                        </select>
                        </td>
                </tr>

                <tr>
                        <td>Languages Known:</td>
                        <td>
                                <input type="checkbox" name="english"
value="English">English<br>
                                <input type="checkbox" name="telugu"
value="Telugu">Telugu<br>
                                <input type="checkbox" name="hindi"
value="Hindi">Hindi<br>
                                <input type="checkbox" name="tamil"
value="Tamil">Tamil<br>
                                </td>
```

```
                                </tr>

                <tr>
                        <td>Address:</td>
                        <td>
                                <textarea name="address" rows="4"
cols="25"></textarea>
                        </td>
                </tr>

                <tr>
                        <td colspan="2" align="left">
                                                <button type="submit"
name="submit" value="Submit">Submit</button>
                                                <button type="reset"
name="reset" value="Reset">Reset</button>
                                                </td>
                </tr>
        </table>
</form>


</body>
</html>
```

S.No: 2	Exp. Name: ***Ravi's Portfolio Page***	**Date: 2026-08-28**

Aim:

Ravi is designing his personal portfolio webpage. He wants each section to look visually distinct. Style the text in each section as described below.

- In the `<header>`, the `<h1>` should have `font-family: Verdana`, `color: royalblue`, `text-shadow: 3px 3px 8px gray`, and `text-align: center`.
- In the `<main>`, the `<h2>` should have `text-transform: uppercase`, `color: darkred`, and `letter-spacing: 2px`.
- In the `<main>`, the `<p>` should have `font-size: 16px`, `color: #333`, and `line-height: 1.7`.
- In the `<footer>`, the `<p>` should have `text-align: center`, `font-style: italic`, and `color: darkgreen`.

Source Code:

index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8" />
  <meta name="viewport" content="width=device-width, initial-
scale=1.0"/>
  <title>Ravi’s Portfolio Page</title>
  <link rel="stylesheet" href="styles.css" />
</head>
<body>
  <header>
    <h1>Welcome to My Portfolio</h1>
  </header>

  <main>
    <h2>about me</h2>
    <p>Hello! I'm Ravi, a passionate front-end developer who loves
designing simple and elegant web experiences for users.</p>
  </main>

  <footer>
    <p>© 2025 Ravi Sharma | Designed with creativity and code.</p>
  </footer>
</body>
</html>
```

styles.css

```
/*Write your code here*/

header h1{
        font-family:verdana;
        color:royalblue;
        text-shadow:3px 3px 8px gray;
        text-align:center
}

main h2{
        text-transform:uppercase;
        color:darkred;
        letter-spacing:2px
}

main p{
        font-size:16px;
        color:#333;
        line-height:1.7
}

footer p{
        text-align:center;
        font-style:italic;
        color:darkgreen
}
```

S.No: 3	Exp. Name: ***Simple Calculator***	**Date: 2026-09-09**

Aim:

You are provided with an HTML file that defines the structure of a basic calculator. The calculator supports the following arithmetic operations:

- Addition (+)
- Subtraction (-)
- Multiplication (*)
- Division (/)

Your task is to complete the JavaScript file (script.js) so that the calculator functions correctly.

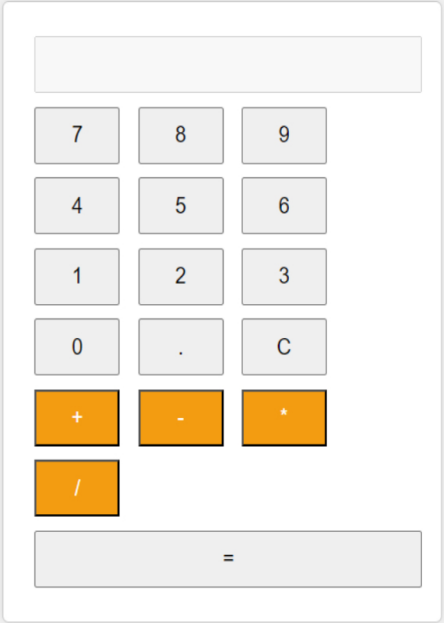
Provided Files

HTML File (index.html)

The HTML file already includes:

- A display area to show the current input or result.
- Buttons for digits (0–9).
- A decimal point button (.).
- Operation buttons (+, −, *, /).
- An equals button (=).
- A clear button (C).

You do not need to modify the HTML file.



Complete the JavaScript Functionality:

You must complete the logic in **script.js** by implementing the following functions correctly:

1. appendNumber(number)

This function is called when a number button is clicked.
It should:

- Allow multi-digit number input.
- Append the clicked digit to the current input.
- Handle decimal input properly (prevent multiple decimals in a single number).

2. setOperation(operation)

This function is called when an arithmetic operator (+, −, *, /) is clicked.
It should:

- Store the current input as the previous operand.
- Store the selected operation.
- Prepare the calculator to receive the next number.
- Prevent setting an operation if there is no current input.

3. calculate()

This function is called when the equals (=) button is clicked.
It should:

- Perform the selected arithmetic operation using the stored previous value and current value.
- Display the result.
- Handle division by zero by displaying:

```
Cannot divide by zero
```

If either the previous value or current value is empty, display:

```
Error
```

4. clearDisplay()

This function is called when the clear (C) button is clicked.
It should:

- Reset the current input.
- Reset the previous input.
- Reset the selected operation.
- Clear the display.

Source Code:

index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-
scale=1.0">
    <title>Simple Calculator</title>
    <style>
        body {
            font-family: Arial, sans-serif;
            display: flex;
            justify-content: center;
            align-items: center;
            height: 100vh;
            background-color: #f4f4f4;
        }
        .calculator {
            border: 1px solid #ccc;
            padding: 20px;
            border-radius: 5px;
            background-color: #fff;
            box-shadow: 0 0 10px rgba(0,0,0,0.1);
        }
        .calculator input, .calculator button {
            margin: 5px;
            padding: 10px;
            font-size: 16px;
        }
        .calculator input {
            width: 100%;
            box-sizing: border-box;
        }
        .calculator button {
            width: 22%;
        }
        .calculator button.operator {
            background-color: #f39c12;
            color: #fff;
        }
    </style>
</head>
<body>
    <div class="calculator">
        <input type="text" id="display" disabled>
```

```
        <br>
        <button onclick="appendNumber('7')">7</button>
        <button onclick="appendNumber('8')">8</button>
        <button onclick="appendNumber('9')">9</button>
        <br>
        <button onclick="appendNumber('4')">4</button>
        <button onclick="appendNumber('5')">5</button>
        <button onclick="appendNumber('6')">6</button>
        <br>
        <button onclick="appendNumber('1')">1</button>
        <button onclick="appendNumber('2')">2</button>
        <button onclick="appendNumber('3')">3</button>
        <br>
        <button onclick="appendNumber('0')">0</button>
        <button onclick="appendNumber('.')">.</button>
        <button onclick="clearDisplay()">C</button>
        <button class="operator" onclick="setOperation('+')">+</button>
        <button class="operator" onclick="setOperation('-')">-</button>
        <button class="operator" onclick="setOperation('*')">*</button>
        <button class="operator" onclick="setOperation('/')">/</button>
        <br>
        <button style="width: 100%;" onclick="calculate()">=</button>
    </div>

    <script src="script.js"></script>
</body>
</html>
```

script.js

```
let currentInput = '';
let previousInput = '';
let operation = null;

function appendNumber(number) {
        currentInput += number;
        updateDisplay();
}

function setOperation(op) {
        if(currentInput === '')
                return;
        if(previousInput!=='')
                calculate();
        operation=op;
        previousInput=currentInput;
        currentInput='';
}

function clearDisplay() {
        currentInput='';
        previousInput='';
        operation=null;
        updateDisplay();
}

function calculate() {
        if(previousInput === '' || currentInput===='' ||
operation===null){
                currentInput='Error';
                updateDisplay();
                return;
        }

        let prev=parseFloat(previousInput);
        let current=parseFloat(currentInput);

        switch(operation)
        {
                case '+':
                        result=prev+current;
                        break;
                case '-':
                        result=prev-current;
```

```
                        break;
                case '*':
                        result=prev*current;
                        break;
                case '/':
                        if(current === 0)
                        {
                                currentInput='Cannot divide by zero';
                                previousInput='';
                                operation=null;
                                updateDisplay();
                                return;
                        }
                        result=prev/current;
                        break;
        }

        currentInput=result.toString();
        previousInput='';
        operation=null;
        updateDisplay();
}

function updateDisplay() {
    document.getElementById('display').value = currentInput;
}

function printToDisplay(message) {
    document.getElementById('display').value = message;
}
```

S.No: 5	Exp. Name: ***Node.js Event Logger Using Packages, Events, Timers, and File I/O***	**Date: 2026-09-30**

Aim:

In this experiment, you will build a Node.js event logging system that connects four core concepts into one working flow. The `server.js`, browser UI, and CSS file are already provided in the default code. Your only task is to implement two things inside `logger.js`.

Your Task

Implement the following inside `logger.js`:

- Create a class **LogEmitter** that extends the built-in `EventEmitter` class.
- Implement the **startLogging** function. It must do the following:
    - Create an instance of **LogEmitter**.
- Register a listener on the `"log"` event. Each time the event fires, the listener must build a log entry in this exact format and append it to `LOG_FILE` using `fs.appendFileSync`:
- `[<uuid>] <timestamp> — <message>\n`
- where `uuid` comes from `uuidv4()` and `timestamp` comes from `new Date().toISOString()`.Use `setInterval` to emit the `"log"` event every `2000` ms with the message `"Event fired"`.Use `clearInterval` to stop the interval after `5` ticks.Emit a `"done"` event on the emitter after the 5th tick.Return the **LogEmitter** instance.

Note: Click **Start** in the browser UI to trigger your implementation and observe the live dashboard. Click **Reset** to clear and run again.

Source Code:

my-app/logger.js

```
const EventEmitter = require("events");
const fs = require("fs");
const path = require("path");
const { v4: uuidv4 } = require("uuid");

const LOG_FILE = path.join(__dirname, "logs", "events.log");

if (!fs.existsSync(path.join(__dirname, "logs"))) {
  fs.mkdirSync(path.join(__dirname, "logs"));
}

// ── TASK 1: Create a custom EventEmitter
──────────────────────────────────

// Create a class LogEmitter that extends EventEmitter.
// Write your code here
class LogEmitter extends EventEmitter {}

// ── TASK 2: Implement the startLogging function
──────────────────────────

// This function must:
//   1. Create an instance of LogEmitter
//   2. Register a listener on the "log" event that:
//        a. Builds a log entry in this exact format:
//           [<uuid>] <timestamp> — <message>\n
//           where uuid comes from uuidv4() and timestamp from new
Date().toISOString()
//        b. Appends the entry to LOG_FILE using fs.appendFileSync
//   3. Use setInterval to emit the "log" event every 2000ms with the
message "Event fired"
//   4. Use clearInterval to stop the interval after 5 ticks
//   5. Emit a "done" event on the emitter after the 5th tick
//   6. Return the LogEmitter instance
function startLogging() {
  // Write your code here
        const emitter=new LogEmitter();
        emitter.on("log",(message)=>{
                const uuid=uuidv4();
                const timestamp=new Date().toISOString();
                const logEntry=`[${uuid}] ${timestamp} — ${message}\n`;
                fs.appendFileSync(LOG_FILE,logEntry);
        });
        let count=0;
        const interval=setInterval(()=>{
                count++;
```

```
                emitter.emit("log","Event fired");
                if(count==5){
                        clearInterval(interval);
                        emitter.emit("done");
                }
        },2000);
        return emitter;
}

module.exports = { startLogging, LOG_FILE };
```

my-app/package.json

```
{
  "name": "my-app",
  "version": "1.0.0",
  "description": "Node.js Events, Timers, Packages and File I/O",
  "main": "server.js",
  "scripts": {
    "start": "node server.js"
  },
  "dependencies": {
    "uuid": "^9.0.0"
  }
}
```

my-app/public/index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8" />
  <meta name="viewport" content="width=device-width, initial-
scale=1.0"/>
  <title>Node.js Event Logger</title>
  <link rel="stylesheet" href="/style.css" />
</head>
<body>

  <div class="page-header">
    <h1>Node.js Event Logger</h1>
    <p class="subtitle">Monitor live event activity powered by your
Node.js implementation</p>
  </div>

  <div class="main">

    <div class="stat-grid">
      <div class="stat-card">
        <div class="stat-label">Status</div>
        <div class="stat-value" id="stat-status">
          <span class="badge idle">Idle</span>
        </div>
      </div>
      <div class="stat-card">
        <div class="stat-label">Ticks Fired</div>
        <div class="stat-value big" id="stat-ticks">0 <span
class="of">/ 5</span></div>
      </div>
      <div class="stat-card">
        <div class="stat-label">Last Event Time</div>
        <div class="stat-value mono" id="stat-time">—</div>
      </div>
      <div class="stat-card">
        <div class="stat-label">Log File Size</div>
        <div class="stat-value mono" id="stat-size">0 bytes</div>
      </div>
    </div>

    <div class="card">
      <div class="card-title">Last Generated UUID</div>
      <div class="uuid-display" id="stat-uuid">—</div>
```

```
    </div>

    <div class="card">
      <div class="card-title">Event Log</div>
      <div class="empty-state" id="empty-state">
        <span class="empty-icon">◯</span>
        <p>No events yet. Click Start to begin.</p>
      </div>
      <div class="log-feed" id="log-feed"></div>
    </div>

    <div class="btn-row">
      <button id="start-btn" onclick="handleStart()">Start</button>
      <button id="reset-btn" class="btn-secondary"
onclick="handleReset()">Reset</button>
    </div>

  </div>

  <script>
    let pollInterval = null;
    let lastCount = 0;

    function setStatus(type, label) {
      document.getElementById("stat-status").innerHTML = '<span
class="badge ' + type + '">' + label + '</span>';
    }

    window.handleStart = async function () {
      document.getElementById("start-btn").disabled = true;
      document.getElementById("reset-btn").disabled = true;
      setStatus("running", "Running");
      try {
        const res = await fetch("/start", { method: "POST" });
        const data = await res.json();
        if (data.status === "error") {
          setStatus("error", "Error");
          document.getElementById("stat-uuid").textContent =
data.message;
          document.getElementById("start-btn").disabled = false;
          document.getElementById("reset-btn").disabled = false;
          return;
        }
        document.getElementById("empty-state").style.display = "none";
        lastCount = 0;
        pollInterval = setInterval(pollLogs, 600);
```

```
      } catch (err) {
        setStatus("error", "Error");
        document.getElementById("start-btn").disabled = false;
        document.getElementById("reset-btn").disabled = false;
      }
    };

    window.handleReset = async function () {
      clearInterval(pollInterval);
      await fetch("/reset", { method: "POST" });
      lastCount = 0;
      document.getElementById("stat-ticks").innerHTML = '0 <span
class="of">/ 5</span>';
      document.getElementById("stat-time").innerHTML = "—";
      document.getElementById("stat-size").textContent = "0 bytes";
      document.getElementById("stat-uuid").innerHTML = "—";
      document.getElementById("log-feed").innerHTML = "";
      document.getElementById("empty-state").style.display = "flex";
      document.getElementById("start-btn").disabled = false;
      document.getElementById("reset-btn").disabled = false;
      setStatus("idle", "Idle");
    };

    async function pollLogs() {
      try {
        const res = await fetch("/logs");
        const data = await res.json();
        const lines = data.lines || [];

        if (lines.length !== lastCount) {
          lastCount = lines.length;
          document.getElementById("stat-ticks").innerHTML =
lines.length + ' <span class="of">/ 5</span>';
          document.getElementById("stat-size").textContent =
data.fileSize + " bytes";

          if (lines.length > 0) {
            const lastLine = lines[lines.length - 1];
            const uuidMatch = lastLine.match(/\[([^\]]+)\]/);
            const timeMatch = lastLine.match(/\]\s+([^\s]+)\s+/);
            if (uuidMatch) document.getElementById("stat-
uuid").textContent = uuidMatch[1];
            if (timeMatch) document.getElementById("stat-
time").textContent = timeMatch[1].replace("T", " ").replace("Z", "
UTC");
          }
```

```
          const feed = document.getElementById("log-feed");
          feed.innerHTML = lines.map(function(line, i) {
            const uuidMatch = line.match(/\[([^\]]+)\]/);
            const timeMatch = line.match(/\]\s+([^\s]+)\s+/);
            const msgMatch  = line.match(/\u2014\s+(.*)/);
            const uuid = uuidMatch ? uuidMatch[1] : "—";
            const time = timeMatch ? timeMatch[1].replace("T", "
").replace("Z", " UTC") : "—";
            const msg  = msgMatch  ? msgMatch[1] : line;
            return '<div class="log-row">'
              + '<span class="log-num">#' + (i + 1) + '</span>'
              + '<div class="log-details">'
              + '<span class="log-msg">' + msg + '</span>'
              + '<span class="log-meta">' + time + '</span>'
              + '<span class="log-uuid">' + uuid + '</span>'
              + '</div></div>';
          }).join("");
          feed.scrollTop = feed.scrollHeight;
        }

        if (!data.isLogging && lines.length >= 5) {
          clearInterval(pollInterval);
          setStatus("done", "Done");
          document.getElementById("start-btn").disabled = false;
          document.getElementById("reset-btn").disabled = false;
        }

        if (!data.isLogging && lines.length === 0) {
          clearInterval(pollInterval);
          setStatus("error", "Error");
          document.getElementById("start-btn").disabled = false;
          document.getElementById("reset-btn").disabled = false;
        }

      } catch (err) {
        clearInterval(pollInterval);
      }
    }
  </script>

</body>
</html>
```

my-app/public/style.css

```
* { box-sizing: border-box; margin: 0; padding: 0; }

body {
  font-family: "Segoe UI", sans-serif;
  background: #f0f2f5;
  min-height: 100vh;
  color: #1a202c;
}

.page-header {
  background: #fff;
  border-bottom: 1px solid #e2e8f0;
  padding: 24px 40px;
  text-align: center;
}

.page-header h1 {
  font-size: 1.4rem;
  font-weight: 700;
  color: #1a202c;
  margin-bottom: 4px;
}

.subtitle { font-size: 0.83rem; color: #718096; }

.main {
  max-width: 680px;
  margin: 32px auto;
  padding: 0 24px;
  display: flex;
  flex-direction: column;
  gap: 20px;
}

.stat-grid {
  display: grid;
  grid-template-columns: 1fr 1fr;
  gap: 14px;
}

.stat-card {
  background: #fff;
  border: 1px solid #e2e8f0;
  border-radius: 12px;
```

```
  padding: 18px 20px;
}

.stat-label {
  font-size: 0.72rem;
  font-weight: 700;
  text-transform: uppercase;
  letter-spacing: 0.07em;
  color: #a0aec0;
  margin-bottom: 10px;
}

.stat-value { font-size: 0.95rem; font-weight: 600; color: #2d3748; }

.stat-value.big {
  font-size: 1.8rem;
  font-weight: 700;
  color: #1a202c;
  line-height: 1;
}

.stat-value.mono {
  font-family: monospace;
  font-size: 0.82rem;
  color: #4a5568;
  word-break: break-all;
}

.of { font-size: 1rem; color: #a0aec0; font-weight: 500; }

.badge {
  display: inline-block;
  padding: 3px 10px;
  border-radius: 20px;
  font-size: 0.75rem;
  font-weight: 700;
  text-transform: uppercase;
  letter-spacing: 0.05em;
}

.badge.idle    { background: #edf2f7; color: #718096; }
.badge.running { background: #ebf8ff; color: #2b6cb0; border: 1px solid
#90cdf4; }
.badge.done    { background: #f0fff4; color: #276749; border: 1px solid
#9ae6b4; }
.badge.error   { background: #fff5f5; color: #c53030; border: 1px solid
```

```
#feb2b2; }

.card {
  background: #fff;
  border-radius: 12px;
  border: 1px solid #e2e8f0;
  padding: 20px 24px;
}

.card-title {
  font-size: 0.72rem;
  font-weight: 700;
  text-transform: uppercase;
  letter-spacing: 0.07em;
  color: #a0aec0;
  margin-bottom: 12px;
}

.uuid-display {
  font-family: monospace;
  font-size: 0.88rem;
  color: #3182ce;
  font-weight: 600;
  background: #ebf8ff;
  border-radius: 7px;
  padding: 10px 14px;
  word-break: break-all;
  letter-spacing: 0.03em;
}

.empty-state {
  display: flex;
  flex-direction: column;
  align-items: center;
  justify-content: center;
  gap: 10px;
  padding: 32px 20px;
  text-align: center;
}

.empty-icon { font-size: 2rem; color: #e2e8f0; line-height: 1; }
.empty-state p { font-size: 0.83rem; color: #a0aec0; }

.log-feed {
  display: flex;
  flex-direction: column;
```

```
  gap: 8px;
  max-height: 280px;
  overflow-y: auto;
}

.log-row {
  display: flex;
  align-items: flex-start;
  gap: 12px;
  background: #f7fafc;
  border-radius: 8px;
  padding: 10px 14px;
  border-left: 3px solid #4299e1;
}

.log-num {
  font-size: 0.72rem;
  font-weight: 700;
  color: #a0aec0;
  min-width: 24px;
  padding-top: 2px;
}

.log-details {
  display: flex;
  flex-direction: column;
  gap: 3px;
  flex: 1;
}

.log-msg { font-size: 0.88rem; font-weight: 600; color: #2d3748; }
.log-meta { font-size: 0.75rem; font-family: monospace; color: #718096;
}
.log-uuid { font-size: 0.72rem; font-family: monospace; color: #a0aec0;
word-break: break-all; }

.btn-row { display: flex; gap: 12px; }

button {
  flex: 1;
  padding: 11px;
  background: #3182ce;
  color: #fff;
  border: none;
  border-radius: 8px;
  font-size: 0.9rem;
```

```
  font-weight: 600;
  cursor: pointer;
  transition: background 0.2s, opacity 0.2s;
}

button:hover   { background: #2b6cb0; }
button:disabled { opacity: 0.5; cursor: not-allowed; }

.btn-secondary {
  background: #fff;
  color: #4a5568;
  border: 1.5px solid #e2e8f0;
}

.btn-secondary:hover { background: #f7fafc; }
```

my-app/server.js

```
const http = require("http");
const fs = require("fs");
const path = require("path");

const PORT = 3000;

const SERVER_FILE = path.join(__dirname, "logger.js");
let lastMtime = fs.statSync(SERVER_FILE).mtimeMs;
setInterval(() => {
  try {
    const mtime = fs.statSync(SERVER_FILE).mtimeMs;
    if (mtime !== lastMtime) {
      console.log("logger.js changed — restarting...");
      process.exit(1);
    }
  } catch (e) {}
}, 1000);

function serveStatic(res, filePath, contentType) {
  fs.readFile(filePath, (err, data) => {
    if (err) {
      res.writeHead(404, { "Content-Type": "text/plain" });
      res.end("404 Not Found");
      return;
    }
    res.writeHead(200, { "Content-Type": contentType });
    res.end(data);
  });
}

let isLogging = false;

function handleRequest(req, res) {
  if (req.method === "GET" && req.url === "/") {
    serveStatic(res, path.join(__dirname, "public", "index.html"),
"text/html");
    return;
  }

  if (req.method === "GET" && req.url === "/style.css") {
    serveStatic(res, path.join(__dirname, "public", "style.css"),
"text/css");
    return;
  }
```

```
  if (req.method === "POST" && req.url === "/start") {
    if (isLogging) {
      res.writeHead(200, { "Content-Type": "application/json" });
      res.end(JSON.stringify({ status: "already running" }));
      return;
    }
    try {
      delete require.cache[require.resolve("./logger")];
      const { startLogging, LOG_FILE } = require("./logger");
      if (typeof startLogging !== "function") {
        res.writeHead(500, { "Content-Type": "application/json" });
        res.end(JSON.stringify({ status: "error", message:
"startLogging is not implemented yet." }));
        return;
      }
      const logDir = path.join(__dirname, "logs");
      if (!fs.existsSync(logDir)) fs.mkdirSync(logDir);
      fs.writeFileSync(LOG_FILE, "");
      isLogging = true;
      const emitter = startLogging();
      emitter.on("done", () => { isLogging = false; });
      res.writeHead(200, { "Content-Type": "application/json" });
      res.end(JSON.stringify({ status: "started" }));
    } catch (err) {
      isLogging = false;
      res.writeHead(500, { "Content-Type": "application/json" });
      res.end(JSON.stringify({ status: "error", message: err.message
}));
    }
    return;
  }

  if (req.method === "POST" && req.url === "/reset") {
    isLogging = false;
    try {
      delete require.cache[require.resolve("./logger")];
      const { LOG_FILE } = require("./logger");
      if (fs.existsSync(LOG_FILE)) fs.writeFileSync(LOG_FILE, "");
    } catch (e) {}
    res.writeHead(200, { "Content-Type": "application/json" });
    res.end(JSON.stringify({ status: "reset" }));
    return;
  }

  if (req.method === "GET" && req.url === "/logs") {
```

```
    try {
      delete require.cache[require.resolve("./logger")];
      const { LOG_FILE } = require("./logger");
      const content = fs.existsSync(LOG_FILE) ?
fs.readFileSync(LOG_FILE, "utf8") : "";
      const lines = content.trim().split("\n").filter(Boolean);
      const fileSize = fs.existsSync(LOG_FILE) ?
fs.statSync(LOG_FILE).size : 0;
      res.writeHead(200, { "Content-Type": "application/json" });
      res.end(JSON.stringify({ lines, isLogging, fileSize }));
    } catch (err) {
      res.writeHead(200, { "Content-Type": "application/json" });
      res.end(JSON.stringify({ lines: [], isLogging: false, fileSize: 0
}));
    }
    return;
  }

  res.writeHead(404, { "Content-Type": "text/plain" });
  res.end("Not Found");
}

const server = http.createServer(handleRequest);
server.listen(PORT, () => {
  console.log(`Server running at http://localhost:${PORT}`);
});
```

S.No: 7	Exp. Name: ***Creating and Nesting Function and Class Components***	**Date: 2026-09-30**

ID: A24126552168

ANIL NEERUKONDA INSTITUTE OF TECHNOLOGY & SCIENCES

Aim:

You are developing a React application for a digital library. The application should display information about a book by creating and nesting React components. In this activity, the React application setup has already been provided. Your task is to create the **Book** function component and the **App** class component and nest the **Book** component inside the **App** component.

Your Task

1. Inside **Book.jsx**, create a function component named **Book** that returns a `<div>` with the class **book-card**, containing:
    - An `<h2>` element with the text "The Great Gatsby".
    - A paragraph element with the text "Author: F. Scott Fitzgerald".

Export the **Book** component.

2. In the **App.jsx** file:
    - Create a class component named **App** that extends **Component**.
    - Import the `Book` component from `./Book`.
    - Inside the `render()` method, display an `<h3>` element with the text "Books".
    - Create a `<div>` with the class name **library**, and nest the `<Book />` component inside it.
    - Return the JSX from the `render()` method.
    - Export the **App** component as the default export.

Expected Output:

When the application is executed, it should display:

```
Books
--------------------------------
The Great Gatsby
Author: F. Scott Fitzgerald
--------------------------------
```

where the **Book** component is nested inside the **App** component.

Note:

- Both **App.jsx** and **Book.jsx** are in the same path.
- Initially, the page will show a placeholder until the required code is completed.

Source Code:

functionAndClassComponents/src/App.jsx

```
import { Component } from "react";
// Import the Book component here.
import Book from "./Book";
import ErrorBoundary from "./components/ErrorBoundary";

// Create the App class component here...
class App extends Component{
        render(){
                return (
                        <div>
                                <h3>Books</h3>
                                <div className="library">
                                        <Book />
                                </div>
                        </div>
                );
        }
}

export default App;
```

functionAndClassComponents/src/Book.jsx

```
// Create the Book function component here...
function Book(){
        return (
                <div className="book-card">
                        <h2>The Great Gatsby</h2>
                        <p>Author: F. Scott Fitzgerald</p>
                </div>
        );
}

export default Book;
```